GLOBAL DRIVE

Data-Driven Car Import Selection & Profitability Analysis

Project Overview
Business concept, rationale, problem statement, objectives, beneficiaries, and project scope.

Target Market: Egyptian Automotive Market

Project Type: End-to-End Data Analytics and Business Intelligence

Prepared as a graduation-project concept. Company operations and transactions are fictional/simulated unless explicitly supported by verified sources.

1. What Is the Project Idea?

Global Drive is a data analytics and business intelligence project designed to help a fictional car-import company operating in Egypt make more informed decisions about which passenger-car models to import, in what quantities, and under which financial conditions.

The project brings together market-demand indicators, competitor listing prices, purchase prices, exchange rates, shipping and insurance costs, applicable import charges, sales, and inventory information. These inputs are analyzed to estimate landed cost, gross profit, gross margin, inventory risk, and the potential impact of changing market or cost conditions.

The final output will be an interactive Power BI dashboard supported by data preparation, SQL analysis, Python analytics, documented calculations, and scenario-based import planning. Recommendations will be based on available evidence and clearly stated assumptions rather than choosing winning models in advance.

2. Why Are We Building This Project?

Car-import decisions involve multiple financial and market factors. A high selling price does not automatically mean a high profit, and a strong margin does not guarantee sufficient demand. Global Drive aims to turn scattered data into a consistent decision-support process.

- Support evidence-based decisions instead of relying only on intuition or isolated price comparisons.
- Make import costs and expected profitability easier to compare across car models.
- Show how exchange-rate movements, shipping-cost changes, and selling-price changes can affect financial results.
- Reduce the risk of tying up too much capital in slow-moving inventory.
- Help management compare alternatives and allocate a limited import budget more transparently.

3. What Problem Does It Solve?

The core problem is the lack of a unified, data-driven method for evaluating car models before import orders are placed. Decision-makers may need to compare data from different sources, account for changing costs, and balance profitability against demand and inventory risk.

Challenge	How Global Drive Addresses It
Unclear market demand	Compares available market-sales data and clearly labeled demand indicators or proxies.
Different costs and currencies	Combines purchase price, exchange rate, freight, insurance, and applicable import expenses into a documented landed-cost calculation.
Misleading price or margin comparisons	Compares expected selling revenue, gross profit, and gross margin together, where the required inputs are available.
Inventory and capital risk	Highlights inventory levels, slow-moving models, turnover, and days in inventory when supported by data.
Limited import budget	Evaluates model rankings and possible import quantities under a stated budget and documented constraints.

4. Project Objectives

- Collect and organize relevant automotive market data and document each source and its limitations.
- Clean, validate, and standardize datasets using Excel and Power Query.
- Build a relational SQL Server database with documented table relationships and data-quality checks.
- Use SQL and Python to explore market patterns, compare prices, calculate costs, and evaluate profitability.
- Create transparent model-ranking criteria using available demand, profitability, competitiveness, and risk indicators.
- Test scenarios such as exchange-rate increases, higher shipping costs, or lower selling prices.
- Develop a budget-allocation approach that respects the available budget and other documented constraints.
- Build a five-page Power BI report covering executive performance, market and demand, cost and profitability, inventory and risk, and import planning.
- Deliver clear, evidence-based recommendations and explain the assumptions and limitations behind them.

5. Who Benefits from the Project?

Beneficiary	Expected Benefit
Company management	A consolidated view of performance, profitability, risk, and budget scenarios to support import decisions.
Purchasing and import teams	A structured way to compare models, suppliers, purchase costs, and import quantities.
Finance and accounting teams	More transparent cost breakdowns, currency assumptions, gross-profit calculations, and sensitivity analysis.
Sales and inventory teams	Better visibility into sales patterns, inventory levels, and potential slow-moving stock.
Data analysts and project evaluators	A practical demonstration of data cleaning, SQL, Python, BI modeling, visualization, and analytical communication.

6. Project Scope: What Is Included?

The following activities are included in the planned project:

- Data sourcing and documentation using suitable public automotive, exchange-rate, and import-cost references where available.
- Creation of clearly labeled simulated company data when genuine operational records are unavailable.
- Data cleaning, validation, transformation, and exploratory analysis.
- SQL Server database design, data loading, quality checks, joins, analytical queries, and reporting views.
- Python analysis for market exploration, cost and profitability calculations, sensitivity analysis, ranking, and budget optimization where the data supports it.
- Power BI data modeling, DAX measures, interactive visuals, filters, tooltips, and navigation across five dashboard pages.
- Documented assumptions, formulas, data limitations, validation checks, and business recommendations.
- A final project presentation explaining the business problem, methodology, findings, and recommendations.

7. What Is Outside the Project Scope?

The project will not attempt to do the following:

- Operate a real car-import business, place actual orders, or execute purchases and sales.
- Provide legal, customs, tax, or investment advice; applicable rules must be verified for the relevant vehicle and import circumstances.
- Guarantee future demand, selling prices, profit, or the success of any import recommendation.
- Present simulated transactions or assumptions as genuine company records or verified market statistics.
- Claim exact model-level demand when the source only provides broader brand-level or listing-based data.
- Build a live production system connected to company ERP, dealer, customs, or supplier systems.

- Automate real-time web scraping or maintain a continuously updated commercial market database unless separately approved and implemented.
- Use machine-learning forecasting when adequate historical data and a justified predictive use case are not available.
- Make final import decisions without human review, commercial judgment, and verification of current costs and regulations.

8. Main Deliverables

Deliverable	Purpose
Business case and documentation	Defines the problem, scope, data sources, methodology, assumptions, and limitations.
Prepared datasets	Contains source data and cleaned data, with simulated records clearly separated.
SQL Server database and scripts	Stores related data and supports quality checks and business analysis.
Python notebooks	Documents cleaning, exploration, profitability, sensitivity, ranking, and optimization work.
Power BI report	Presents the analysis through five interactive dashboard pages.
Recommendations and presentation	Communicates supported findings, possible import scenarios, and next steps.

9. Success Criteria

- All important metrics have documented formulas and traceable input data.
- Real, simulated, and derived data are clearly distinguished.
- Database relationships and calculations are checked to reduce duplication and double-counting.
- Dashboard KPIs are reconciled against SQL or Python calculations where possible.
- The report communicates the trade-offs between demand, profitability, budget, and risk.
- Recommendations clearly state their evidence base, assumptions, and limitations.

Important limitation
Global Drive is a decision-support prototype. Its recommendations are only as reliable as the available data, cost assumptions, and constraints. All simulated company records must be labeled as simulated, and current import duties and taxes must be verified from appropriate official sources.